CANDIDATE NAME	

CLASS		INDEX NUMBER	

CHEMISTRY 9729/04

Paper 4 Practical

15 Aug 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, name and class on all the work you hand in.
Give details of the practical shift and laboratory when appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 18 and 19.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/ **13**
2	/ **23**
3	/ **19**
Total	/ **55**

This document consists of **19** printed pages and **1** blank page.

1 In this experiment you will determine the relative atomic mass, A_r, of magnesium by a titration method.

FA 1 is 2.00 mol dm^{-3} hydrochloric acid, HC*l*.
FA 3 is 0.120 mol dm^{-3} sodium hydroxide, NaOH
magnesium ribbon
bromophenol blue indicator

(a) Method

Reaction of magnesium with FA 1

- Pipette 25.0 cm^3 of **FA 1** into the 250 cm^3 beaker.
- Weigh the strip of magnesium ribbon and record its mass.

mass of magnesium = ……………g

- Coil the strip of magnesium ribbon loosely and then add it to the **FA 1** in the beaker.
- Stir the mixture occasionally and wait until the reaction has finished.

Dilution of the excess acid

- Transfer all the solution from the beaker into the volumetric flask.
- Make up the solution up to the mark using distilled water.
- Shake the flask to mix the solution before using it for your titrations.
- Label this solution of hydrochloric acid **FA 2**.

Titration

- Fill the burette with **FA 2**.
- Rinse the pipette out thoroughly. Then pipette 25.0 cm^3 of **FA 3** into a conical flask.
- Add several drops of bromophenol blue indicator.
- Run **FA 2** from the burette into this flask until the mixture just becomes yellow.
- Carry out as many titrations as you think necessary to obtain consistent results.
- Make certain that your recorded results show the precision of your working.
- Record in a suitable form below all your burette readings and the volume of **FA 2** added in each accurate titration.

Results

M1	
M2	
M3	
M4	
M5	
M6	

[6]

(b) From your titrations, obtain a suitable volume of **FA 2** to be used in your calculations. Show clearly how you obtained this volume.

M7	

Volume of **FA 2** = ……………………………..[1]

(c) Calculations

Show your working and appropriate significant figures in the final answer to **each** step of your calculations.

(i) Deduce the amount of hydrochloric acid in the volume of **FA 2** you calculated in **(b)**

M8	

Amount of HC*l* = ………………… [1]

(ii) In **(a)**, you reacted 25.0 cm^3 of **FA 1** with your weighed piece of magnesium. After the reaction, the unreacted hydrochloric acid was used to prepare 250 cm^3 of **FA 2**.

Calculate the amount of hydrochloric acid that reacted with the magnesium ribbon.

M9	
M10	

Amount of HC*l* reacting with Mg = ……………………[2]

(iii) Hence, calculate the relative atomic mass , A_r, of magnesium.

M11	

A_r of Mg = ……………………[1]

(d) A student carried out the same experiment but used 1.00 g of magnesium ribbon. State and explain why the student's experiment could not be used to determine the value for the A_r of magnesium.
Include a calculation in your answer.

[A_r: Mg, 24.3]

M12	
M13	

…………………………………………………………………………………………………..

……………………………………………………………………………………………………[2]

[Total: 13]

2 You will investigate the rate of reaction between iron(III) ions, Fe^{3+}, and iodide ions, I^-.

$$2Fe^{3+}(aq) + 2I^{-}(aq) \longrightarrow 2Fe^{2+}(aq) + I_2(aq)$$

The iodine, I_2, produced can be reacted immediately with thiosulfate ions, $S_2O_3^{2-}$.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \longrightarrow 2I^{-}(aq) + S_4O_6^{2-}(aq)$$

When all the thiosulfate has been used, the iodine produced will turn starch indicator blue-black.

The rate of the reaction can therefore be measured by finding the time for the blue-black colour to appear.

FA 4 is aqueous iron(III) chloride, $FeCl_3$.
FA 5 is aqueous potassium iodide, KI.
FA 6 is 0.0060 mol dm^{-3} sodium thiosulfate, $Na_2S_2O_3$.
starch indicator

You are advised to read the instructions before starting any practical work and draw a table for your results in the space on page **6**.

(a) Method

Experiment 1

- Fill a burette with **FA 4**.
- Run 20.00 cm^3 of **FA 4** into a 100 cm^3 beaker.
- Use the measuring cylinder to place the following in a second 100 cm^3 beaker.
  - 10 cm^3 of **FA 5**
  - 20 cm^3 of **FA 6**
  - 10 cm^3 of starch indicator
- Add the contents of the second beaker to the first beaker and start timing.
- Stir the mixture once and place the beaker on the white tile.
- The mixture turns brown and then yellow before turning a blue-black colour. Stop timing when this **blue-black colour** appears.
- Record in your table the volume of **FA 4** used, the volume of distilled water used and the time to the **nearest second** for the blue-black colour to appear.
- Wash both beakers.

For each of **Experiments 1 - 6** you should complete your results table to show the volume of **FA 4** used, the volume of distilled water used and the time taken to the **nearest second** for the blue-black colour to appear.

Experiment 2

- Fill the other burette with distilled water.
- Run 10.00 cm^3 of **FA 4** into a 100 cm^3 beaker.
- Run 10.00 cm^3 of distilled water into the same beaker.
- Use the measuring cylinder to place the following in a second 100 cm^3 beaker.
  - 10 cm^3 of **FA 5**
  - 20 cm^3 of **FA 6**
  - 10 cm^3 of starch indicator
- Add the contents of the second beaker to the first beaker and start timing.
- Stir the mixture once and place the beaker on the white tile.
- Stop timing when a blue-black colour appears.

- Wash both beakers.

Experiments 3 - 6

Carry out **four** further experiments to investigate the effect of changing the concentration of Fe^{3+} (aq) by altering the volume of aqueous $FeCl_3$, **FA 4**, used.

You should not use a volume of **FA 4** that is less than 6.00 cm^3 and the total volume of the reaction mixture must always be 60 cm^3.

M14	
M15	
M16	
M17	
M18	
M19	

[6]

Calculations

The rate of reaction can be found by calculating the change in concentration of Fe^{3+}(aq) that occurred when enough iodine was produced to change the colour of the indicator to blue-black.

Use your data and the equations on page **5** to carry out the following calculations.

Show your working and appropriate significant figures in the final answer to **each** step of your calculations.

(b) **(i)** Calculate the amount of thiosulfate ions, $S_2O_3^{2-}$ used in each experiment.

M20	

Amount of $S_2O_3^{2-}$ ……………………….[1]

(ii) Calculate the amount of iron(III) ions, Fe^{3+}, that were used to produce the amount of iodine that react with the amount of $S_2O_3^{2-}$ in **(i)**.

M21	

Amount of Fe^{3+} ………………………. [1]

(iii) Using your answer to **(ii)**, calculate the change in Fe^{3+} up to the time of appearance of the blue black colour.

M22	

Change in concentration of Fe^{3+} ……………………….[1]

(iv) The following formula can be used as a measure of the 'rate of reaction'.

$$\text{'rate of reaction'} = \frac{\text{change in concentration of } Fe^{3+}(aq)}{\text{reaction time}} \times 10^{6}$$

Complete the table to show the volume of **FA 4**, the reaction time and the rate in **Experiments 1- 6**. You should include units.

If you were unable to calculate a value for the change in concentration of Fe^{3+}(aq) in **(iv)**, you should assume it is 2.50×10^{-3} mol dm^{-3}. (Note: this is not the correct value.)

Experiment			
1			
2			
3			
4			
5			
6			

M23

[1]

(c) On the grid, plot the rate (*y*-axis) against the volume of **FA 4** (*x*-axis). Draw a line of best fit through the points. You should identify any points you consider anomalous.

M24	
M25	
M26	

[3]

(d) Using your graph, what conclusion can you reach about the effect of changing the concentration of $FeCl_3$ on the rate of the reaction between Fe^{3+}(aq) and I^-(aq)?

……

…… | M27 | |

……[2] | M28 | |

(e) It was found, by carrying out experiments similar to those used in **(a)**, that increasing the concentration of I^- increased the rate of the reaction.

The student suggested a modification to the method by using the same volumes of all reagents but with the concentration of **FA4** and **FA 5** being doubled their original values. State what the effect would be on the **reaction time** in Experiment 1 and explain how this change would affect any **possible errors** in the measurements.

……

……

……[1] | M29 | |

(f) Which of the experiments you carried out in **(a)** had the greatest percentage error in the reaction time? Calculate this percentage error. Assume that the error in measuring the reaction time is ±0.5 s.

[1] | M30 | |

(g) Planning

Aqueous hydrogen peroxide decomposes into oxygen gas and water. The reaction is normally very slow but is catalysed by solid manganese (IV) oxide.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

You are to plan an experiment to investigate how the rate of the catalysed decomposition of aqueous hydrogen peroxide depends on its concentration.

(i) The rate of decomposition depends on the number of hydrogen peroxide molecules present in a given volume of solution.

Use this information to predict how the rate of decomposition of the hydrogen peroxide depends on the concentration.

Prediction…………………………………………………………………………………

………………………………………………………………………………………….

…………………………………………………………………………………………[1]

M31

(ii) You are to design a laboratory experiment to test your prediction in **(g) (i)**.

In addition to the standard apparatus present in a laboratory you are provided with the following materials.

- 0.1 mol dm^{-3} aqueous hydrogen peroxide
- A supply of manganese (IV) oxide

Complete the table below to show how you would prepare five solutions of aqueous hydrogen peroxide. Make sure that the correct units are recorded.

expt No.	volume of H_2O_2	volume of H_2O	concentration of H_2O_2
1			
2			
3			
4			
5			

M32

[1]

(iii) Give a step by step description of how you would carry out **one** complete experiment.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[3]

M33	
M34	
M35	

(iv) State a problem which might be experienced by someone having to carry out these experiments alone.

…………………………………………………………………………………………………..

…………………………………………………………………………………………………[1]

M36	

[Total: 23]

3 Qualitative Analysis

At each stage of any test you are to record details of the following:

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitates in an excess of the reagent added

Where gases are released they should be identified by a test, **described in the appropriate place in your observations.**

You should indicate clearly at which stage of the test a change occurs.
Marks are **not** given for chemical equations.
No additional tests for ions present should be attempted.

If any solution is warmed, a boiling tube MUST be used.

Rinse and reuse test-tubes and boiling tubes where possible.

Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.

(a) **FA 7** and **FA 8** are mixtures. Each mixture contains one cation and two anions from those listed on pages 18 and 19.

Carry out the following tests and record all your observations in the table below.

Test	Observations	
	FA 7	**FA 8**
(i) Place two spatula measure of **FA 7** into a boiling tube, then add approximately 5 cm^3 of dilute nitric acid using the measuring cylinder. You are to repeat the steps for **FA 8**. Then		
add about 5 cm^3 of distilled water and shake to mix the solution. Use a 1 cm depth of the solution obtained in a test-tube for each of tests **(ii) – (v)**.	No observation required.	
(ii) Add aqueous sodium hydroxide. Then add dilute nitric acid dropwise until in excess.		Do not carry out this test.
(iii) Add aqueous ammonia.		Do not carry out this test.

Test	Observations	
	FA7	**FA8**
(iv) Add aqueous barium nitrate followed by dilute nitric acid.		

M37	

	FA7	FA8
(v) Add aqueous silver nitrate followed by aqueous ammonia.		

M38	
M39	
M40	

[4]

(vi) Use your observations to identify the following ions.

FA 7 contains cation ………………………………….

FA 7 contains anions ………………………………… and ……………………………….
FA 8 contains anions ………………………………   and ……………………………….

State and explain all the evidence for your identification of the cation in **FA 7**.

…………………………………………………………………………………………………

…………………………………………………………………………………………………...

…………………………………………………………………………………………………[3]

M41	
M42	
M43	

(b) **FA 9** contains two cations from the list on page 18.

(i) Transfer approximately half of the **FA 9** into a hard-glass test-tube and heat gently at first, and then strongly, until no further change is seen.
Test with litmus papers while you are heating.
Record all your observations below.

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[1]

M44	

(ii) Carry out further tests that will enable you to identify **both** cations in **FA 9**.
Describe your tests briefly and state your observations.

FA 9 contains cations ………………………….……… and ……………………………….…
[2]

M45	
M46	

(c) Planning

A solder is an alloy of metals which is used to join other metal pieces together.

A specialist solder that can be used to join together pieces of aluminium is made from a mixture by mass of 65% zinc, 20% aluminium and 15% copper.

You are to plan an experimental procedure to confirm the composition of a powdered sample of this solder, by adding reagents and then extracting from the mixture each of the following in sequence;

(i) the copper metal,
(ii) the aluminium as aluminium hydroxide,
(iii) the zinc as zinc hydroxide

You are provided with

- a sample of this solder, with approximate mass 4 g,
- 1.00 mol dm^{-3} sulfuric acid,
- 1.00 mol dm^{-3} ammonia

No other reagents should be used. Standard laboratory equipment is available including a balance, accurate to two decimal places.

(i) Complete the flowchart below to show the order in which the reagents would be added to the solder to allow you to extract and separate the components as copper metal, (**Step 1**), aluminium hydroxide, **(Step 2)**, and zinc hydroxide, **(Step 3)**.
You are reminded that aqueous ammonia contains both the base OH^- and the complex-forming molecule NH_3.

Step 1	**Step 2**	**Step 3**
reagent(s) added	reagent(s) added	reagent(s) added
………………………..........	…………………………	……………………………
…………………………....	…………………………	……………………………....
substance(s) present at the end of the reaction	substance(s) present at the end of the reaction	substance(s) present at the end of the reaction
…………………………....	………………………..........	……………………………
…………………………....	………………………..........	……………………………
…………………………....	……………………………	……………………………
substance(s) removed by filtration (if any)	substance(s) removed by filtration (if any)	substance(s) removed by filtration (if any)
…………………………....	……………………………	……………………………
…………………………....	……………………………	……………………………

M47	
M48	
M49	
M50	
M51	

[5]

(ii) For some of the steps in the procedure you would need to be careful to add an appropriate quantity of a reagent.
For each step of your procedure explain why particular quantities of reagent should be chosen.

Step 1

…………………………………………………………………………………………………

………………………………………………………………………………………………… M52

…………………………………………………………………………………………………[1]

Step 2

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[1] M53

Step 3

…………………………………………………………………………………………………

…………………………………………………………………………………………………

…………………………………………………………………………………………………[1] M54

(iii) If the mass of aluminium hydroxide obtained was 1.50 g, calculate the mass of aluminium that was present in the solder.
[A_r: A*l*, 27.0; O, 16.0]

[1] M55

[Total: 19]

Qualitative Analysis Notes
[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	*reaction with*	
	NaOH(aq)	**NH_3(aq)**
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^{+}(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white. ppt. with high [Ca^{2+}(aq)]	no ppt.
chromium(III), Cr^{3+}(aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu^{2+}(aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt. turning brown on contact with air insoluble in excess	green ppt. turning brown on contact with air insoluble in excess
iron(III), Fe^{3+}(aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off-white ppt. rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

ion	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^- (aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^- (aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^- (aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^- (aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil
nitrite, NO_2^- (aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil; NO liberated by dilute acids (colourless NO → (pale) brown NO_2 in air)
sulfate, SO_4^{2-} (aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-} (aq)	SO_2 liberated on warming with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

(c) Tests for gases

gas	*test and test result*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	“pops” with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

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